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Controllable contour and colour morphology for privacy-usability trade-offs in thumbnail-preserving encryption

Yijie Lin [a], Ching-Chun Chang [b], Chin-Chen Chang [a,*], Jui-Chuan Liu [a]

[a] Department of Information Engineering and Computer Science, Feng Chia University, Taichung, 407, Taiwan
[b] National Institute of Informatics, Tokyo, 101-8430, Japan

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ABSTRACT

With the proliferation of cloud-based multimedia storage, secure and user-friendly visual interfaces are essential for addressing the inherent tension between data privacy and usability. Traditional encryption protects data but eliminates meaningful visual previews. While existing thumbnail-preserving encryption (TPE) methods attempt to bridge this gap, they often suffer from rigid configurations, typically relying solely on most-significant-bit (MSB)-based contour preservation while neglecting colour semantics. This inflexibility severely limits both visual adaptability and privacy control. To address this issue, this paper proposes a novel controllable contour and colour morphology TPE scheme that functions as a highly adaptable visual privacy interface. Specifically, cryptography-guided morphing with dynamically adjustable contour and colour weights is introduced, allowing users to customize the desired trade-off between visual recognizability and content confidentiality. Experimental results, validated through a subjective user survey, demonstrate that jointly modelling contour and colour features not only enhances perceptual usability but also strengthens protection against visual leakage. Furthermore, the proposed scheme achieves full reversibility with zero storage overhead, while its visual quality can be dynamically configured, with PSNR values ranging from approximately 8 to 32 dB, to accommodate diverse application requirements. These characteristics highlight the substantial potential of C&C-TPE to serve as a configurable, secure, and standardizable visual interface for modern cloud-based multimedia management.

1. Introduction

In the era of data explosion, massive amounts of data are transmitted and stored in cloud environments through various channels. The growing popularity of social media and the rapid adoption of mobile devices, such as smartphones and other connected devices, have further facilitated the storage of large image collections in the cloud and the instant sharing of photographs. However, the increasing reliance on cloud storage and remote image access raises significant concerns regarding data security and privacy. In particular, protecting image content while retaining useful previews for browsing and management has become an important practical requirement, highlighting the need for effective image protection mechanisms that balance security and accessibility.

Data encryption is essential for preserving privacy and managing information, particularly in cloud service applications. Encrypting an image involves using an encryption key to generate a cipher image [1]. To enable secure communications, several encryption schemes have been developed [2–4]. For mobile devices like smartphones and tablets, maintaining consistent file sizes after data hiding is crucial to ensure transmission speed and optimize the user experience. In cloud storage scenarios commonly associated with mobile environments, downloading and decrypting all images to locate a specific one can be inefficient and time-consuming. To address this issue, low-resolution thumbnail images have been introduced. These thumbnails enable users to quickly identify targeted images without the need to download or decrypt all stored content, thereby improving access efficiency and reducing network bandwidth usage.

Unlike traditional thumbnails, thumbnail-preserving encryption (TPE) [5,6] is designed not only to enable quick transmission but also to protect the original image while maintaining usability. As a result, the size of the TPE image remains the same as the original image. Fig. 1 illustrates TPE's application in cloud service scenarios. In Fig. 1(a), an original image is stored on the cloud server. Users can preview and understand the content, but if hackers access the file, they can view it as well. This demonstrates high usability but poor privacy. In contrast,

* Corresponding author.
E-mail addresses: p1263670@o365.fcu.edu.tw (Y. Lin), ccchang@nii.ac.jp (C.-C. Chang), ccc@o365.fcu.edu.tw (C.-C. Chang), jaluanno@gmail.com (J.-C. Liu).

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Fig. 1(c) shows an encrypted image where attackers cannot identify the content, but users are also unable to preview it. This demonstrates strong privacy but low usability. To address this trade-off, many TPE methods [7–21] have been proposed. As shown in Fig. 1(b), the original image is processed to generate a TPE image, which is then stored on the cloud server. With prior knowledge, users can reconstruct the original image from the TPE image to ensure usability. However, attackers who lack prior knowledge cannot interpret the content, ensuring privacy. TPE, therefore, emerges as an effective technology in cloud environments, balancing usability and privacy.

Data hiding [22–30] conceals secret information within a medium, such as an image. Reversible data hiding (RDH) [31–34] refers to techniques that allow the stego image to be completely restored to the original image. Reversible data hiding in encrypted images (RDHEI) [35–38] comprises methods that do not increase file size when embedding secret data into encrypted images. To meet the image privacy and usability needs of cloud-based multimedia applications, we propose a novel customised contour-colour morphing in thumbnail-preserving encryption (C&C-TPE) scheme that compresses the image into blocks to embed recovery information and data, combines the contour and colour features of the image, assigns weights, and generates a morphing TPE image. The TPE images can be controlled through weight assignments based on the specific application scenarios, while the freed space can be used to embed secret data. For example, when browsing a personal cloud album, a larger contour weight may help users recognise familiar outlines while reducing the prominence of original colours. A larger colour weight may be preferable when dominant colours help distinguish scenes. By parameterizing the morphing guidance weights, this scheme can be seamlessly integrated into existing cloud storage systems as a standardized privacy-preserving interface protocol. The contributions of this paper are:

- TPE images are reversible to their original form, meeting the demand for lossless image recovery in cloud storage applications.
- TPE images maintain the same file size as the original images, avoiding additional storage overhead.
- A controllable contour-colour morphology mechanism is introduced, which allocates weights between contour and colour information in TPE images to balance privacy and usability in cloud storage applications. This explicit weighting allows the same image to have contour-oriented or colour-oriented previews, giving users control over which visual cues are retained.
- TPE images can embed secret data in the freed space.

The rest of the paper is structured as follows: Section 2 discusses related studies; Section 3 describes the details of the proposed scheme; Section 4 presents the experimental results and their analysis; the final conclusion and future study are completed in Section 5.

2. Related Work

Preventing information leakage from thumbnail images is critical for maintaining privacy. Thumbnail-preserving encryption is a concept proposed to protect privacy and maintain the prior information of the thumbnails for usability. Wright et al. [7] proposed a method in 2015 to exchange pixels within a pixel block in an encrypted JPEG image and produce a thumbnail effect. A thumbnail effect means that there are no detailed features shown but a rough sketch. Unless users have a general idea of the image, it is difficult to identify the image to which a thumbnail corresponds. For image owners, it is easy to manage their own images by looking at these rough sketches. Marohn et al. [8], in 2017, proposed TPE-like schemes which can encrypt images kept by thumbnails. Later, a cryptography scheme using thumbnails was proposed by Tajik et al. [9] to balance usability and privacy. Since these schemes fail to restore images completely, Yang et al. [12] proposed a method that can fully restore the original image in 2022.

Fig. 2 shows thumbnail-preserving encrypted images generated by different methods. Fig. 2 (a) shows the original image. Fig. 2 (b) illustrates the high-fidelity scheme of HF-TPE proposed by Zhang et al. in 2021 [10], which is irreversible to the original image. PRA-TPE, proposed by Ye et al. in 2022 [13], is depicted in Fig. 2 (c); it uses difference expansion [39] to adjust pixels and results in a TPE image with

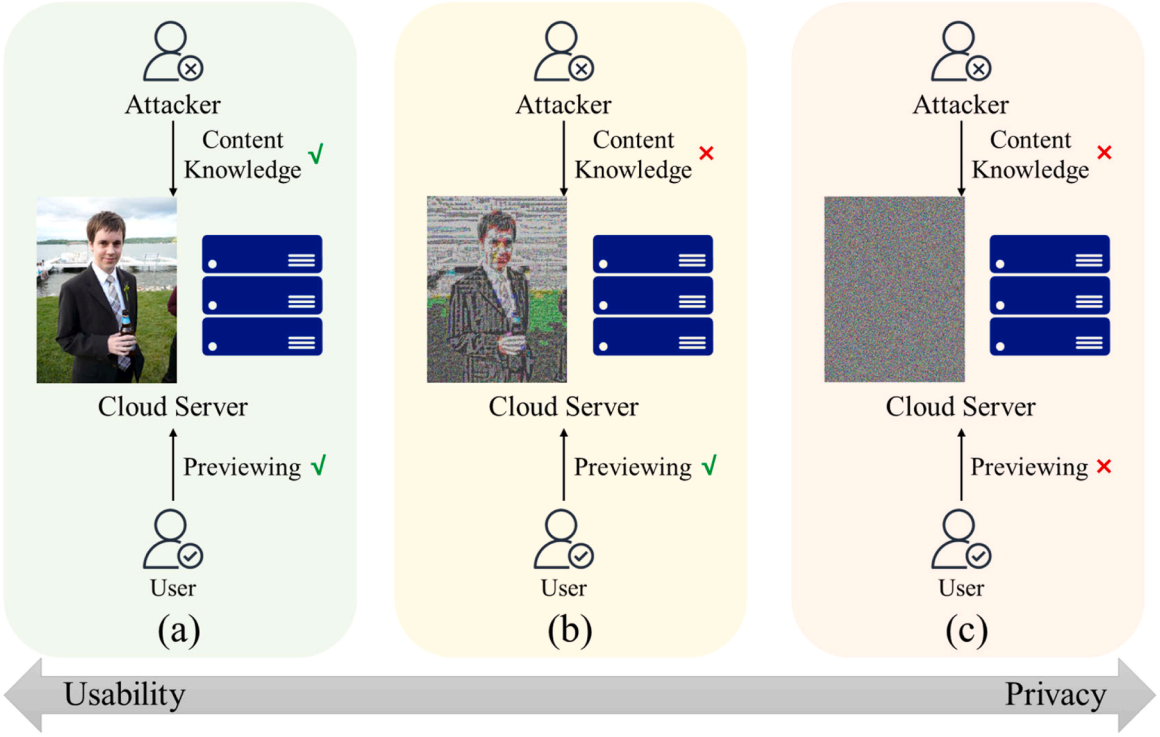

Fig. 1. Cloud service application scenarios between usability and privacy. (a) Original image. (b) TPE image. (c) Encrypted image.

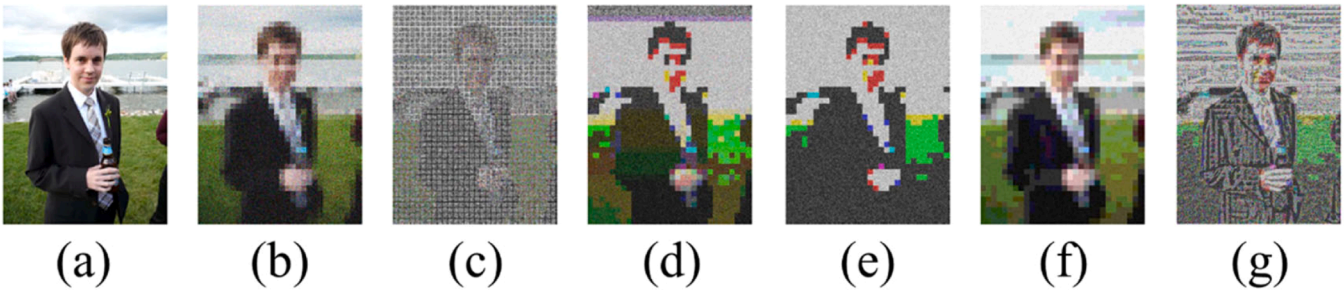

Fig. 2. Thumbnail-preserving encryption images generated by different schemes. (a) Original image. (b) HF-TPE [10]. (c) PRA-TPE [13]. (d) [15]. (e) [16]. (f) TP-HVN [19]. (g) Proposed C&C-TPE.

noticeable block noise. In 2022, Ye et al. [15] utilised the method of Puteaux and Puech [40] to create space for image processing and store auxiliary information. The TPE image of this approach is shown in Fig. 2 (d), where the upper part of the image exhibits visible noise due to the stored auxiliary information. In 2023, Zhang et al. [16] suggested combining different RDHEI schemes and using various embedding methods for different regions of the image, as demonstrated in Fig. 2 (e). Fig. 2 (f) presents TP-HVN, introduced by Wang et al. [19] in 2024 and aimed to produce TPE images resembling traditional thumbnails. Our proposed C&C-TPE utilises the contour and the colour features of the image to generate morphing TPE images by assigning weights, as shown in Fig. 2 (g). In comparison, HF-TPE [10] prioritises thumbnail fidelity without lossless recovery, whereas PRA-TPE [13] and the RDHEI-based methods [15,16,19] support reversibility. Existing work also offers tunable visual usability [16]; C&C-TPE specifically controls the relative contributions of contour and colour information through explicit weights.

3. Proposed C&C-TPE scheme

This section elaborates on the proposed C&C-TPE scheme. Fig. 3 shows the flow of the scheme. The content owner first performs block compression on the original image and stores auxiliary information for image recovery. The freed space can then be used to embed data. Contour and colour weights are subsequently applied for morphing guidance to generate a TPE image, which is stored on a cloud server. Upon receiving the TPE image, the receiver can perform image recovery and data extraction. Table 1 lists the symbols used.

Table 1
List of symbols.

Symbol	Meaning
w_b	Block width
h_b	Block height
m	Block mean
i	Pixel index
p_i	Original pixel value
d_i	Difference between p_i and m
K_e	Encryption key
K_d	Data hiding key
K_r	Random key
$\boldsymbol{OS}_{AI}$	Original stream of auxiliary information
$\boldsymbol{ES}_{AI}$	Encrypted stream of auxiliary information
$\boldsymbol{R}_{AI}$	Random stream generated by K_e
$\boldsymbol{OS}_{SM}$	Original stream of secret message
$\boldsymbol{ES}_{SM}$	Encrypted stream of secret message
$\boldsymbol{R}_{SM}$	Random stream generated by K_d
$contour_i$	Contour bitmap value
$colour_i$	Colour bitmap value
n_{MSB}	Number of MSB retained in colour bitmap
$\boldsymbol{GM}$	Guide map generated by K_r
$\boldsymbol{BM}_{contour}$	Contour bitmap
$\boldsymbol{BM}_{colour}$	Colour bitmap
$\boldsymbol{BM}_{morphing}$	Morphing bitmap generated by $\boldsymbol{GM}$
$\oplus$	XOR operation
$\lfloor\cdot\rceil$	Rounding to nearest integer
$\lfloor\cdot\rfloor$	Rounding down to nearest integer

3.1. Block compression

To compress the original image, we first separate the colour image

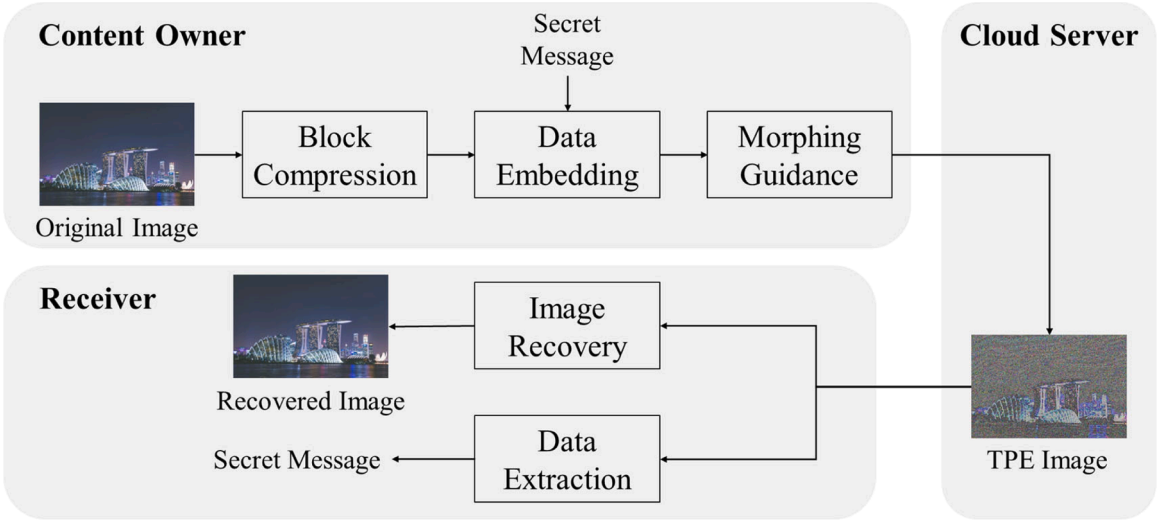

Fig. 3. Flow of C&C-TPE.

into three channels, then divide each channel into blocks of the same size and calculate the average value within each block according to Eq. (1), where m represents the mean of the block, w_b represents the width of the block, h_b represents the height of the block, p_i represents the original pixel value, and i represents the index being processed. Next, we compute the difference between each original pixel value p_i and the mean m as shown in Eq. (2). The mean value is directly assigned to the first pixel of each block, while the remaining differences are collected as auxiliary information. After processing all blocks, Huffman coding [41] is applied to these differences to generate the compressed auxiliary information. For security reasons, we need to encrypt the compressed auxiliary information. The original stream $\boldsymbol{OS}_{AI}$ includes the length of the Huffman coded stream and the Huffman coded stream itself. We first use the encryption key K_e as a seed to generate a pseudo-random stream $\boldsymbol{R}_{AI}$ of the same length as the auxiliary information, and then use the original stream $\boldsymbol{OS}_{AI}$ with a stream cipher to obtain the encrypted stream $\boldsymbol{ES}_{AI}$, as shown in Eq. (3). The $\boldsymbol{ES}_{AI}$ is stored in the vacating position.

$$m = \left\lfloor \frac{\sum_{i=1}^{w_b \times h_b} p_i}{w_b \times h_b} \right\rfloor. \tag{1}$$

$$d_i = p_i - m. \tag{2}$$

$$\boldsymbol{ES}_{AI} = \boldsymbol{OS}_{AI} \oplus \boldsymbol{R}_{AI}. \tag{3}$$

3.2. Data embedding

After block compression, the freed space can be used for data embedding. We use the same encryption method as the auxiliary information to encrypt the secret message. We first use the data hiding key K_d as a seed to generate a pseudo-random stream $\boldsymbol{R}_{SM}$ with the same length as the secret message, and then encrypt the original stream $\boldsymbol{OS}_{SM}$ using the stream cipher to obtain the encrypted stream $\boldsymbol{ES}_{SM}$, as shown in Eq. (4). Finally, $\boldsymbol{ES}_{SM}$ is stored in the freed space corresponding to d in the encoded stream structure.

$$\boldsymbol{ES}_{SM} = \boldsymbol{OS}_{SM} \oplus \boldsymbol{R}_{SM}. \tag{4}$$

3.3. Morphing guidance

Contour and colour are both important components of an image. We aim to use both features and propose C&C-TPE to generate contour and colour morphing for TPE.

First, we generate contour bitmaps and colour bitmaps to guide the generation of the MSBs for the TPE image. In Eq. (5), i represents the index of the pixel currently being processed, $contour_i$ represents the value of the contour bitmap at the position corresponding to i, and p_i represents the original pixel. If the original pixel value is less than or equal to the mean m, 0 is used as the contour bit; otherwise, 1 is recorded. All contour bits in the block form the contour bitmap $\boldsymbol{BM}_{contour}$. In Eq. (6), the colour bit $colour_i$ takes the n_{MSB} bits of the original pixel p_i, where $n_{MSB} \in [1, 4]$ is determined by the 2-bit indicator. The MSBs of all pixels in the block form the colour bitmap $\boldsymbol{BM}_{colour}$. In summary, the number of bits for the contour bitmap is always 1, while the number of MSB retained in colour bitmap, n_{MSB}, is determined by the block's 2-bit indicator and may range from 1 to 4.

$$contour_i = \begin{cases} 0, & if\ p_i \le m \\ 1, & otherwise \end{cases}. \tag{5}$$

$$colour_i = \left\lfloor \frac{p_i}{2^{8-n_{MSB}}} \right\rfloor,\ n_{MSB} \in [1, 4] \tag{6}$$

Then, we need to assign weights to $\boldsymbol{BM}_{contour}$ and $\boldsymbol{BM}_{colour}$ to guide the morphing generation. The weights are set according to the needs of the application scenario, depending on whether the contour features or the colour features are more important. After setting the weights, we will, using a single 32-bit random key K_r as a guided random seed, generate a binary guide map $\boldsymbol{GM}$ for each block based on the weight assignment. In $\boldsymbol{GM}$, 0 represents the contour and 1 represents the colour, and the ratio of 0 to 1 must match the set weight. Each block's $\boldsymbol{GM}$ is guided by $K_r$but exhibits an almost unique distribution, mimicking natural noise. We then generate $\boldsymbol{BM}_{morphing}$ according to the guidance of $\boldsymbol{GM}$. Random generation is preferred over fixed or rule-based positions, as using a predetermined or identical $\boldsymbol{GM}$ for each block may lead to visually discernible patterns. Given the human visual system's sensitivity to repetitive structures, consistent $\boldsymbol{GM}$s can create noticeable lines, blocks, or other regular artifacts. In contrast, random $\boldsymbol{GM}$s mimic natural noise, thereby enhancing the effectiveness of TPE. This effect is illustrated by the fixed and random guide maps in Fig. 9.

Fig. 4 shows an example of morphing guidance with a block size of 4×4 and $n_{MSB} = 1$. First, the mean $m = 27$ of the original block pixels is calculated. Then, $\boldsymbol{BM}_{contour}$ and $\boldsymbol{BM}_{colour}$ are calculated according to Eq. (5) and Eq. (6) respectively. Assume that we set the weights of contour and colour to 0.5 and 0.5, and randomly generate $\boldsymbol{GM}$ to perform morphing guidance. If the bit value at a particular position in the $\boldsymbol{GM}$ is 0, it means that the position in $\boldsymbol{BM}_{morphing}$ is covered by the corresponding $\boldsymbol{BM}_{contour}$. Conversely, if the bit value in the $\boldsymbol{GM}$ is 1, it means that the corresponding position in $\boldsymbol{BM}_{morphing}$ is covered by $\boldsymbol{BM}_{colour}$. After processing all positions, we can then obtain its $\boldsymbol{BM}_{morphing}$.

Finally, for each block in each channel, we assign its $\boldsymbol{BM}_{morphing}$ as the MSBs of the block's pixels, and fill the vacated bits with the encoded stream ES_{AI} to construct the proposed C&C-TPE image.

3.4. Image recovery and data extraction

After receiving the TPE image, the receiver first uses the random key K_r to generate the $\boldsymbol{BM}_{morphing}$ for all blocks, and then extracts the encoded stream. Next, the receiver reads the length of the auxiliary information, which can be used to calculate the length of the secret message. If the receiver possesses the encryption key K_e, it is used as a seed to generate a pseudo-random stream $\boldsymbol{R}_{AI}$ of the same length as the auxiliary information, and the auxiliary information is then decrypted according to Eq. (7). The decrypted auxiliary information is then Huffman decoded to obtain the pixel differences. Using Eq. (8), the original pixel p_i can be recovered. After processing all pixels, the original image is restored. If the receiver has the data hiding key K_d, it is used as a seed to generate a pseudo-random stream $\boldsymbol{R}_{SM}$ of the same length as the secret message. The secret message can then be decrypted according to Eq. (9) and extracted. If the receiver has both K_e and K_d, both the original image and the secret message can be recovered. In summary, the random key controls guide-map generation, the encryption key protects the auxiliary information for image recovery, and the data hiding key protects the secret message.

$$\boldsymbol{OS}_{AI} = \boldsymbol{ES}_{AI} \oplus \boldsymbol{R}_{AI}. \tag{7}$$

$$p_i = d_i + m. \tag{8}$$

$$\boldsymbol{OS}_{SM} = \boldsymbol{ES}_{SM} \oplus \boldsymbol{R}_{SM}. \tag{9}$$

4. Experimental results

In this section, we evaluated the performance of the proposed C&C-TPE scheme using 400 test images from the DIV2K dataset [42]. The experiments were conducted on a Windows 11 laptop with an AMD Ryzen 7 5800H CPU, 16 GB of RAM, and MATLAB 2024a.

Fig. 5 shows the TPE images generated by our proposed scheme. The first row displays the original image, while the second to fourth rows show TPE morphing images using different block sizes. The columns represent different weights, with changes from contour to colour from left to right.

We use the peak signal-to-noise ratio (PSNR) and the structural

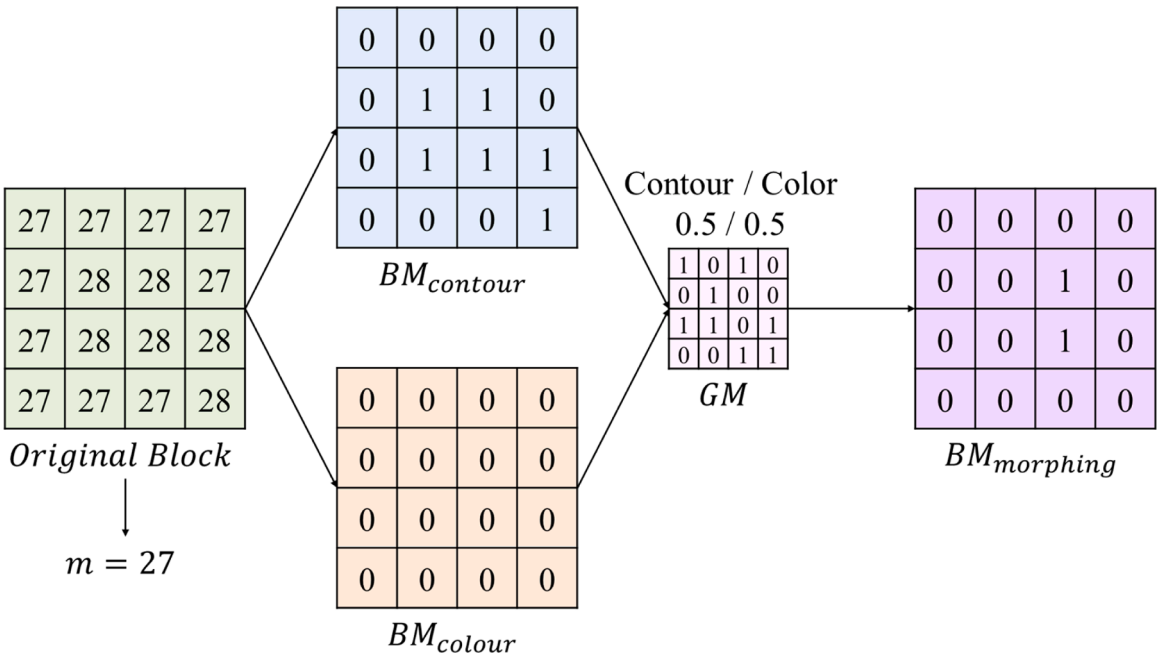

Fig. 4. An example of morphing guidance. The contour and colour bitmaps provide the two feature sources, the guide map selects between them, and the morphing bitmap records the selected result.

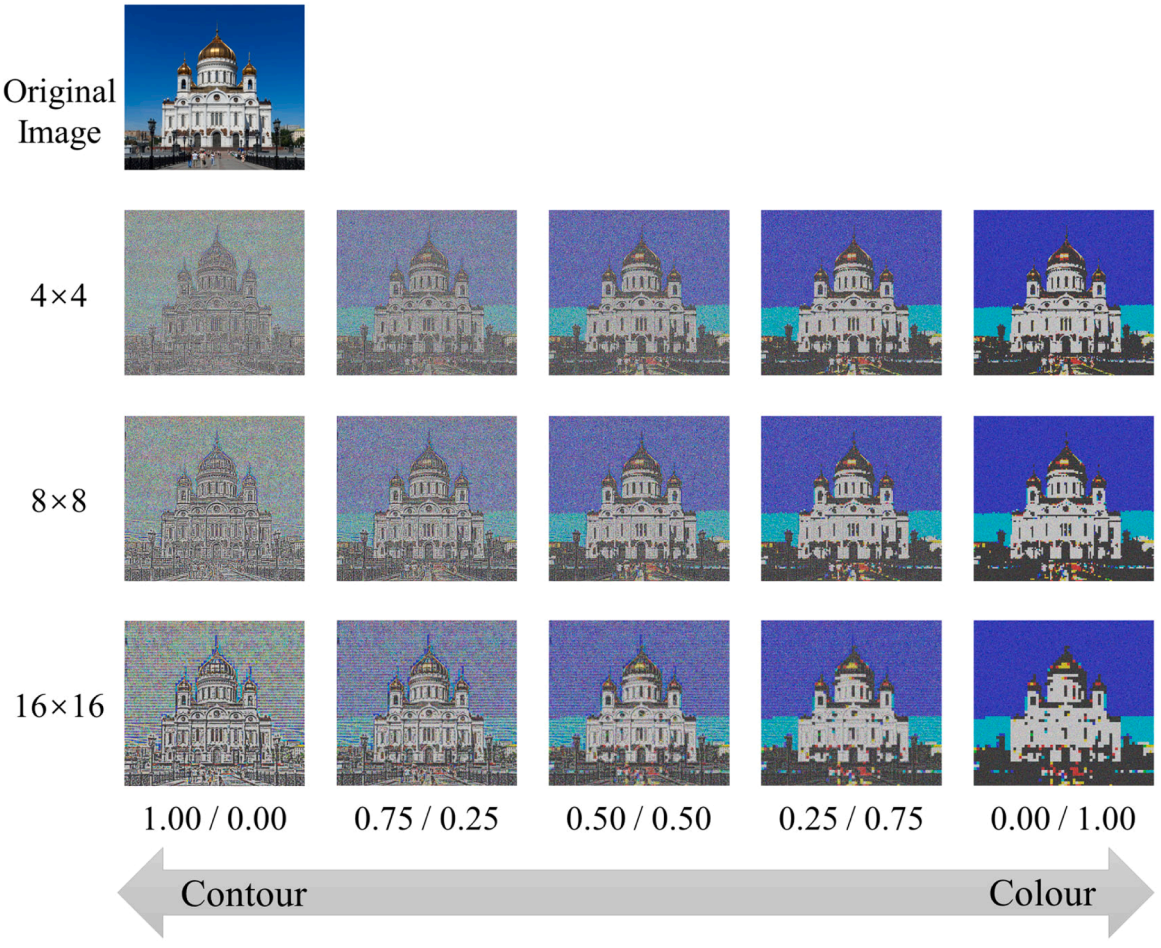

Fig. 5. TPE images with different block sizes and weights.

similarity index (SSIM) as indicators to measure visual quality. A higher PSNR indicates greater similarity between the two images, and an SSIM

closer to 1 signifies that their structures are more alike. Fig. 6 shows the performance of TPE images with different weights for a block size of 16 × 16. The numbers below each TPE plot represent PSNR and SSIM, respectively. We observe that, for all images, a larger contour weight results in a lower PSNR, which aligns with our expectations. This is because using the contour bit as the MSB of the TPE pixel causes a significant change in the pixel value. Additionally, SSIM values are close to 0, indicating little similarity to the original images.

Since the visual perception of TPE images is highly subjective, we invited 30 volunteers to simulate user preferences by voting on multiple sets of TPE images with different weight configurations. No standardised guidance on balancing usability and privacy was provided; volunteers made their selections according to their own subjective criteria. Two scenarios were designed: one allowing a single selection and the other allowing multiple selections. The number of times each image was selected was recorded, and percentages were calculated based on these counts. Table 2 shows the results of the subjective survey. In the single-selection scenario, most volunteers chose contour and colour weights of 0.5 each, with up to 67% making this choice. Similarly, in the multiple-selection scenario, the 0.5 and 0.5 weights also had the highest proportion, reaching 50%. This indicates that most volunteers believe contour and colour are equally important for TPE images. When the contour weight or colour weight was set to 1, almost no volunteers chose these options, further confirming this preference. Apart from the 0.5 and 0.5 weights, in the single-selection scenario, 20% of volunteers chosen weights closer to contour, while 13% desired weights closer to colour. In the multiple-selection scenario, 26% favored contour, and 24% preferred colour. This suggests that, except for the case where both weights are 0.5, most volunteers consider contour to be slightly more important than colour.

Table 2
Subjective survey on contour and colour.

Contour / Colour	1.00 / 0.00	0.75 / 0.25	0.50 / 0.50	0.25 / 0.75	0.00 / 1.00
Single Choice	0%	20%	67%	10%	3%
Multiple Choice	2%	24%	50%	18%	6%

Based on the conclusions from the subjective survey in Table 2, we set both the contour weight and the colour weight to 0.5 and conducted experiments with different block sizes. Fig. 7 presents the performance of TPE images for various block sizes. The numbers below each TPE plot represent PSNR and SSIM, respectively. For the same image, the differences in PSNR and SSIM across block sizes are not substantial. This indicates that block size has a minimal effect on the visual quality of TPE images when using balanced weights.

Fig. 8 shows TPE images with different weights and values of n_{MSB} for a block size of 4 × 4. The two metrics used to evaluate the TPE images are the PSNR and SSIM between the TPE images and the original images. It can be observed that a larger n_{MSB} results in colour information closer to that of the original image, along with higher PSNR and SSIM values, reaching up to 27.8201 dB and 0.9493, respectively. We do not explicitly define whether the PSNR and SSIM values represent the effectiveness of TPE, as this is a highly subjective judgment for users. In other words, we provide customizable options that allow users to choose different block sizes, weights, and n_{MSB} values, thereby demonstrating the flexibility and effectiveness of our scheme. For a particular application, equal weights can serve as an initial setting based on the survey preference. Users can increase the contour weight when familiar

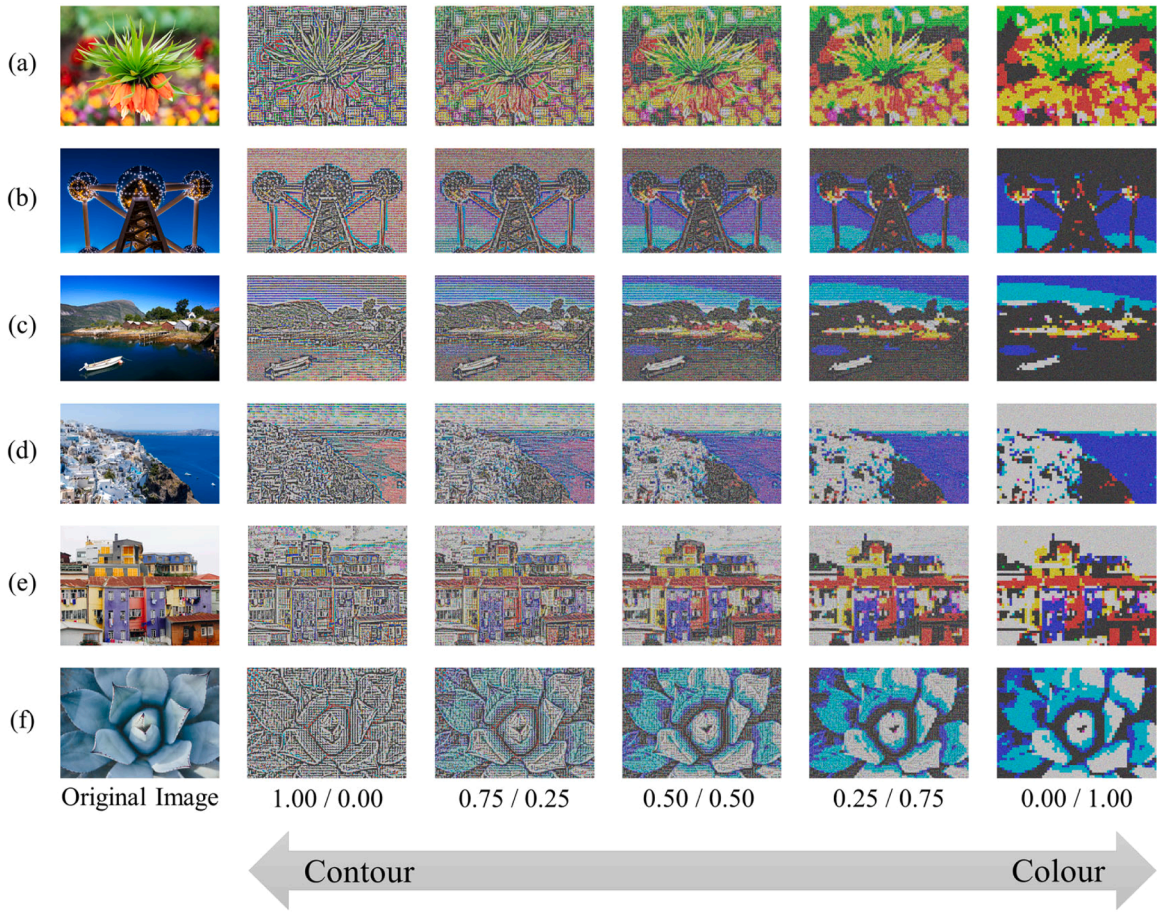

Fig. 6. TPE images with different weights for a block size of 16 × 16, and $n_{MSB} = 1$.

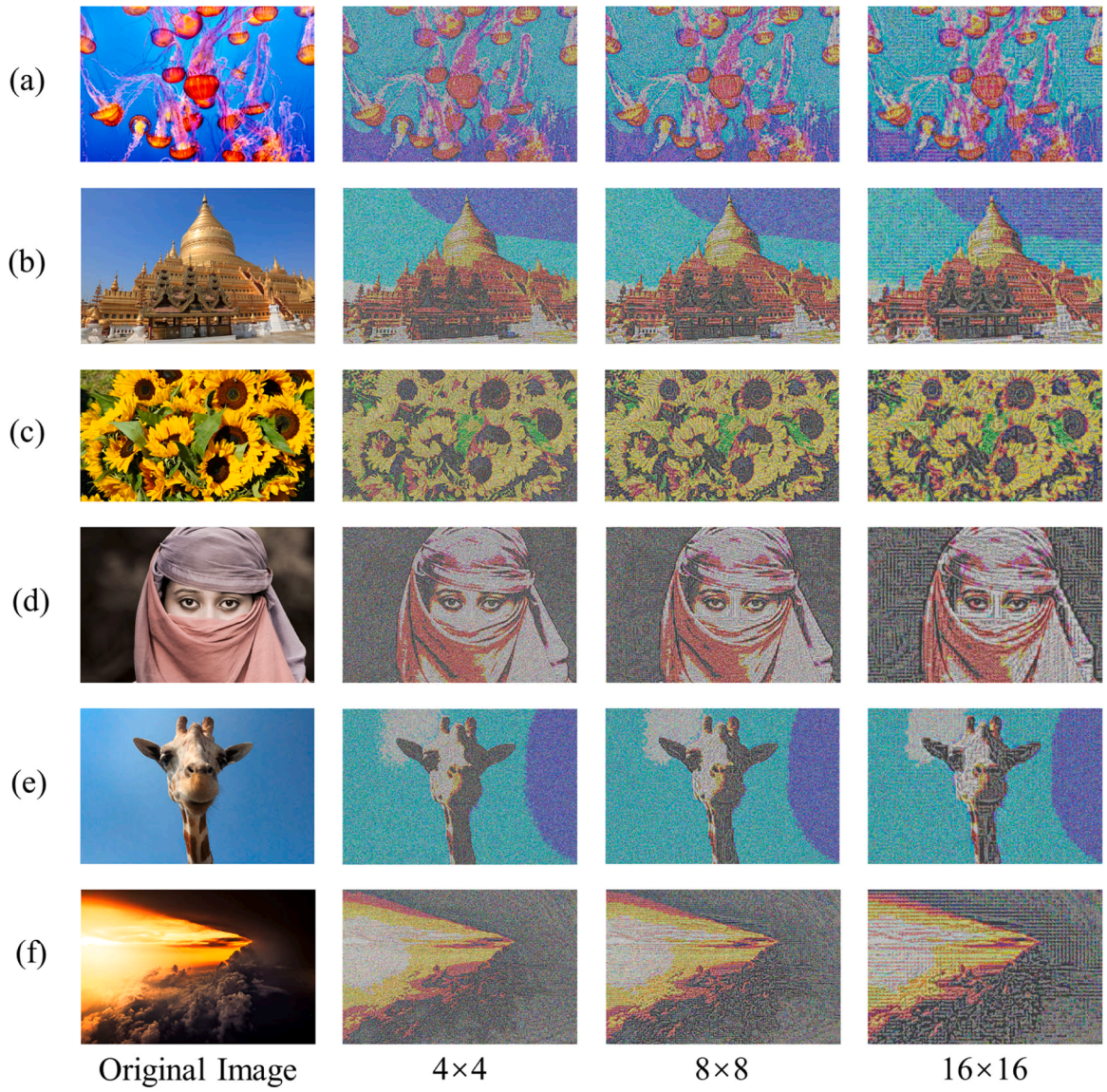

Fig. 7. TPE images with different block sizes when the contour weight and the colour weight are both set to 0.5, and $n_{MSB} = 1$.

outlines aid recognition, or the colour weight when colour cues help distinguish images. Candidate settings should be checked on representative images against the intended browsing task and acceptable visual disclosure.

Fig. 9 shows the results of TPE images generated using different guide maps. Fig. 9(a) corresponds to a guide map with a fixed horizontal average distribution, while Fig. 9(b) corresponds to a guide map with a fixed vertical average distribution. It can be observed that the stripes in Fig. 9(a) are horizontal, whereas those in Fig. 9(b) are vertical. This visual artifact is pronounced. Fig. 9(c) shows the proposed random guide map, which exhibits no obvious stripes but appears similar to noise. These results demonstrate that the random guide map effectively eliminates directional artifacts, improving the visual quality of TPE images.

We attempted to simulate the attacker's perspective by assuming that several plaintext-ciphertext pairs were leaked. Fig. 10 shows the images recovered using the known-plaintext attack with different weights and $n_{MSB} = 1$. As can be seen, the attack fails. This is because the actual encrypted portion has no strong correlation with the known plaintext, making it impossible to derive the key or recover sensitive details. The large key space and inherent randomness ensure security, rendering the scheme resistant to the known-plaintext attack.

Table 3 shows the visual performance of TPE images generated from the DIV2K dataset [42]. It can be observed that the average PSNR is close to 10.6 and the average SSIM is close to 0.16 for all block sizes. It confirms our statement that block size has little effect on weight balance. The main difference lies in the payload which refers to bits per pixel (bpp). On average, the smaller the block size, the larger the payload. When the block size is 4×4, the average payload reaches 6.7198 bpp.

To further evaluate the quantitative performance, Table 4 compares the TPE images and their resulting thumbnails with the original thumbnail. Here, the Thumbnail Images are obtained by directly performing block compression on the original images, whereas the Resulting Thumbnails (RT) are obtained by compressing the corresponding TPE images. The compared irreversible schemes [9–11,14] generally yield low similarity for TPE images relative to the original thumbnails, while their RT images achieve higher similarity. The proposed C&C-TPE scheme is evaluated under three parameter settings: Set 1 with contour weight = 1.0 and colour weight = 0.0 (n_{MSB} is irrelevant); Set 2 with contour weight = 0.5, colour weight = 0.5, and $n_{MSB} = 1$; and Set 3 with contour weight = 0.5, colour weight = 0.5, and $n_{MSB} = 4$. Sets 1 and 2, which emphasize contour information, result in TPE images that are less similar to the original thumbnails, whereas their RT images maintain moderate similarity. Set 3 generates a TPE image exhibiting visual characteristics more similar to the thumbnail, thereby illustrating

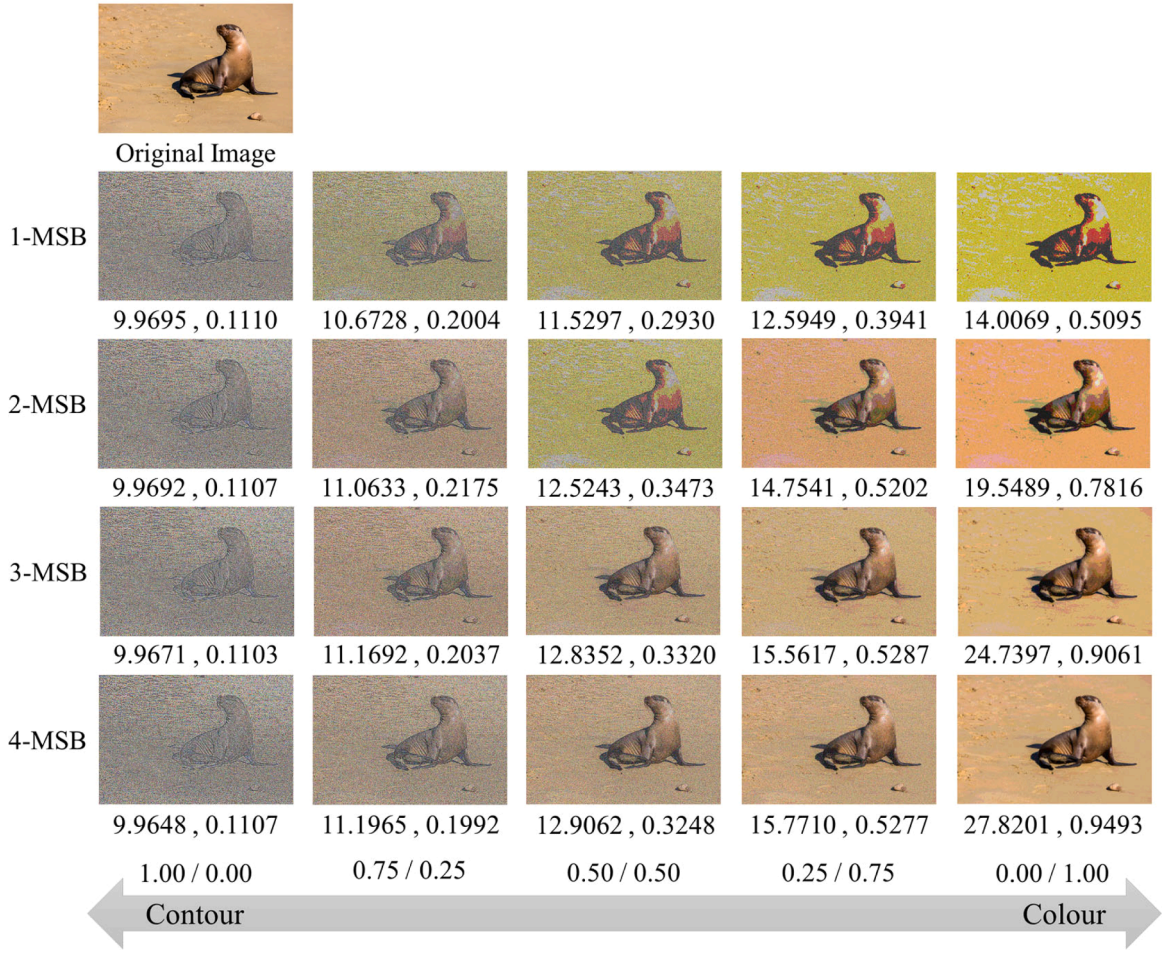

Fig. 8. TPE images with different weights and n_{MSB} for a block size of 4 × 4.

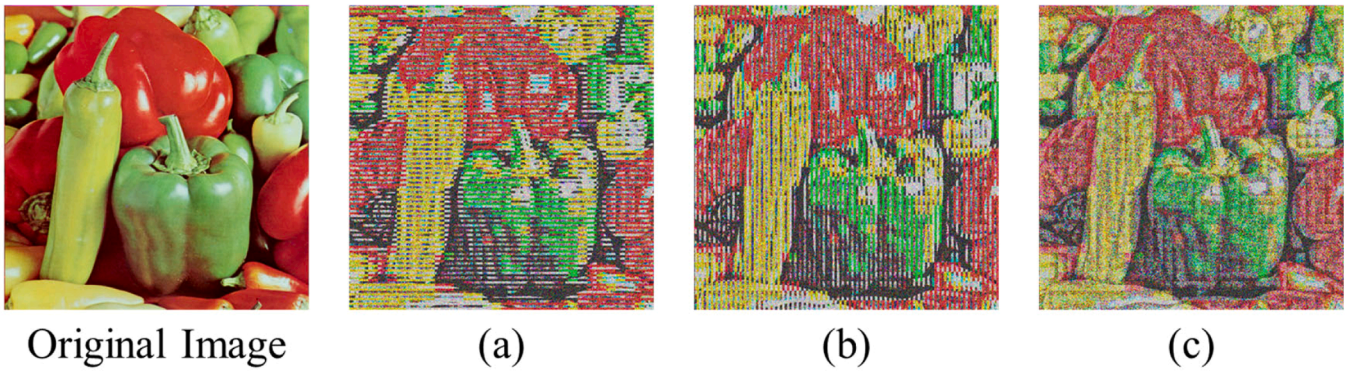

Fig. 9. TPE images using different guide maps.

the influence of different parameter settings on the trade-off between visual fidelity and contour preservation.

As shown in Table 5, sum-preserving encryption (SPE) schemes often exhibit limited reversibility or low visual quality in the generated thumbnails. In contrast, RDHEI-based schemes, including the proposed method as well as those in [15,16], and [19], consistently achieve reversibility while providing higher or adjustable visual quality. Notably, the proposed scheme offers a flexible image quality range (≈8-32 dB) depending on the selected block size, contour and colour weights, and n_{MSB}, demonstrating its adaptability compared with fixed-parameter RDHEI methods. The proposed scheme is classified as RDHEI-based because compressed recovery information and an encrypted payload are embedded within the image. Recovery relies on this auxiliary information rather than preservation of each block's pixel sum, allowing the visible contour-colour representation to be adjusted while retaining lossless recovery.

5. Conclusions

This paper proposes a novel C&C-TPE scheme that integrates both contour and colour features, enabling customisation based on user preferences. By assigning weights to these aspects, the scheme facilitates a controllable trade-off between usability and privacy. Experimental results show that TPE images remain sufficiently recognisable despite

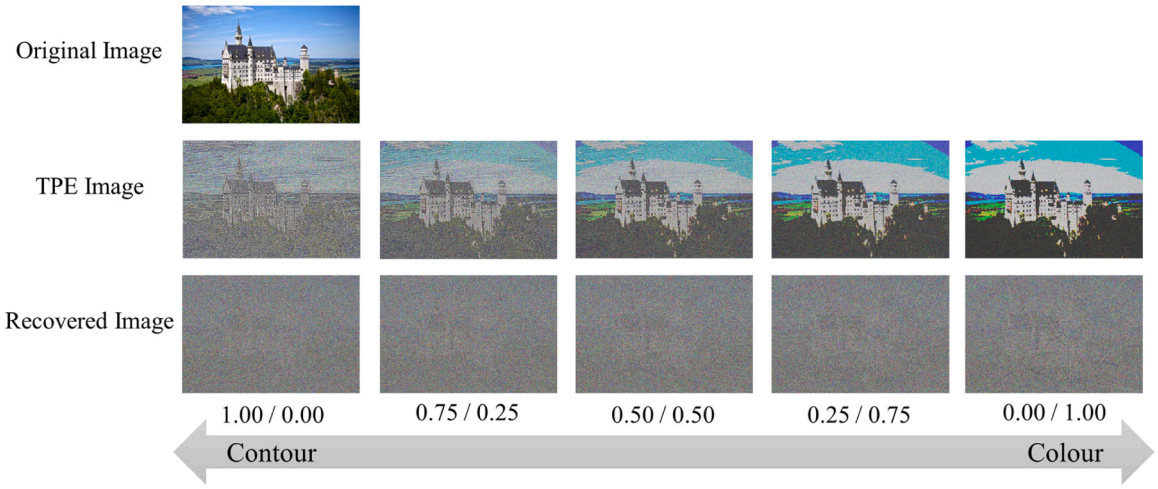

Fig. 10. Recovered image using the known-plaintext attack with different weights.

Table 3
Performance of TPE images generated from 400 test images with different block sizes, with both the contour weight and colour weight set to 0.5, and $n_{MSB} = 1$.

Block Size	Metric	Maximum	Minimum	Average
4 × 4	PSNR	12.2276	6.4782	10.5759
	SSIM	0.4297	0.0210	0.1607
	Payload	15.6503	2.9042	6.7198
8 × 8	PSNR	12.1233	6.2995	10.5482
	SSIM	0.4285	0.0214	0.1593
	Payload	14.6988	1.0211	4.3851
16 × 16	PSNR	12.1300	6.3017	10.6646
	SSIM	0.4425	0.0214	0.1575
	Payload	13.9209	0.0126	2.7862

Table 4
Quantitative comparison with thumbnail.

Schemes		Thumbnail and TPE		Thumbnail and RT	
		PSNR	SSIM	PSNR	SSIM
Irreversible Schemes	[9]	13.2498	0.1654	Inf	1
	[10]	14.1095	0.2457	42.7845	0.9972
	[11]	13.8234	0.1823	Inf	1
	[14]	11.5588	0.1462	32.1896	0.9531
Reversible Schemes	[12]	13.2555	0.206	24.8548	0.8953
	[13]	8.7747	0.0235	2.1376	0.8463
	[15]	15.6409	0.2761	28.4167	0.9349
	[16]	19.316	0.3718	32.9021	0.9643
	[19]	28.3621	0.7493	32.4961	0.9598
	Proposed (Set 1)	8.4126	0.0118	11.8105	0.2203
	Proposed (Set 2)	10.3671	0.0872	15.3658	0.3786
	Proposed (Set 3)	31.7923	0.8229	34.5899	0.9529

Table 5
Overall comparison.

Scheme	Processing Approach	Reversibility	Image Quality
[10]	SPE	No	<15 dB
[13]	SPE	Yes	<10 dB
[15]	RDHEI	Yes	<20 dB
[16]	RDHEI	Yes	<20 dB
[17]	SPE	Yes	<10 dB
[18]	SPE	Yes	<15 dB
[19]	RDHEI	Yes	>28 dB
Proposed	RDHEI	Yes	≈8-32 dB

reduced visual quality, illustrating a practical balance between usability and privacy. A subjective survey suggests that setting both contour and colour weights to 0.5 achieves the most favourable results, with contours generally regarded as more important. This supports the integration of contour information into TPE image generation. While the scheme offers flexibility and reversibility, its performance depends on parameter selection and image characteristics, which may limit universality across diverse datasets. Nevertheless, the findings highlight the potential of C&C-TPE to enhance user privacy while preserving usability, providing a highly adaptable and standardized visual interface solution for modern cloud-based multimedia environments. Future work will focus on optimising adaptation and expanding customisation options.

CRediT authorship contribution statement

Yijie Lin: Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Ching-Chun Chang:** Writing – review & editing, Supervision. **Chin-Chen Chang:** Writing – review & editing, Resources, Conceptualization. **Jui-Chuan Liu:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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